

Two-jump magnetic switching in ultrathin Co(0 0 1) films probed by giant magnetoresistance measurements

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Abstract

Ultrathin epitaxial FCC-Co films, which form part of a spin-valve structure, were found to undergo one- or two-jump magnetic switching, in GMR and MOKE measurements depending upon the field orientation. The transitions are mediated by the propagation of 180° or 90° domain walls. The Co two-jump spin switching in the spin-valve structure has contributed to the formation of three stable GMR states: parallel, antiparallel and a new intermediate state.

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Orientation-dependent magnetic switching behavior of ultrathin magnetic films has been an interesting topic of study for many years. In this case epitaxial Fe films on GaAs system, which generally has an in-plane magnetization, and a combination of strong cubic magnetocrystalline anisotropy and interface-induced uniaxial anisotropy [1–7] is the most studied system. For instance, Florczak et al. [1] have used longitudinal and transverse magneto-optical Kerr effect (MOKE) to study the magnetization reversal in the Fe/GaAs(1 0 0) system and have found that the reversal in ultrathin epitaxial Fe can proceed via an intermediate state, in which the spins are oriented at 90° to the initial and final remanent directions. The observations of Chen and Erskine [2] in the Fe/W(0 0 1) system have revealed that when an in-plane uniaxial anisotropy is intentionally induced in the films, then both classic one-irreversible-switch hysteresis loops, and loops showing two irreversible jumps can be obtained, depending on the applied field orientation. Using Lorentz microscopy, Gu et al. [3] were able to explain how the one- or two-jump switching in the Fe/GaAs(0 0 1) system develops via domain processes. Cowburn et al. [4,5] have observed three irreversible jumps

between four stable spin configurations, and they have also developed a phenomenological model based on anisotropy and domain-wall pinning energy. The energy model described well the dependence of coercive field and the one-, two- or three-jump nature of the switch on the applied field orientation in the Ag/Fe/Ag(0 0 1) system. While many investigations have focused on the orientation-dependent magnetization reversal process of ultrathin epitaxial Fe films, such studies in the Cu/Co system have yet received little attention, despite the Cu/Co system having been investigated intensively [8–13]. In this work, we investigate the orientation-dependent magnetization reversal in epitaxial FCC Co(0 0 1) films using the effect of the giant magnetoresistance, which method has been widely used to probe the domain reversal processes [14,15].

Molecular beam epitaxy techniques were used to grow epitaxial samples in this work. During the growth, in situ reflection high electron diffraction measurements were carried out to confirm epitaxial growth of the FCC-Co films. The trilayer spin-valve sample has a nominal composition of Si(0 0 1)/FCC-Cu(100 Å)/FCC-NiFe(60 Å)/FCC-Cu(60 Å)/FCC-Co(40 Å)/FCC-Cu(50 Å). The Co top magnetic layer is of main interest and the use of a spin-valve structure enables GMR measurement to be used to probe the Co magnetic properties. A standard four-point

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probe was used for the GMR measurements. Longitudinal MOKE measurements were carried out to obtain hysteresis loops. All measurements were done at room temperature. An epitaxial single-layer Co film with a structure of Si(001)/Cu(100 Å)/Co(40 Å)/Cu(50 Å) was also grown for transverse MOKE measurements.

Fig. 1 shows the observed GMR loop with the field applied along the $[\bar{1}10]$ direction, which is also the cubic easy axis. This is a classic GMR loop with both NiFe and Co magnetization parallel to each other ($\theta = 0^\circ$) in saturation, but their magnetizations are in complete antiparallel configuration ($\theta = 180^\circ$) in the range $\pm(10\text{--}50)\text{Oe}$. The parallel configuration gives minimum GMR, while the antiparallel configuration gives maximum GMR. The first magnetic switching at $\sim 10\text{Oe}$ is due to the NiFe film. As the field increases to $H_c \approx 50\text{Oe}$, the Co magnetization switches directly from the initial remanent direction to the final remanent direction through nucleation of 180° reverse domains and a wall propagation process. The 180° Co magnetization transition is termed one-jump Co magnetic switching. In Fig. 1(b), the sample was rotated such that the field was at $\phi = 15^\circ$ to the $[\bar{1}10]$ direction. The GMR loop unexpectedly shows additional plateaux with a height near one-quarter of the value of the GMR change ($\Delta R = R_{\text{antiparallel}} - R_{\text{parallel}}$). The presence of triple switch-

ing in the loop indicates that, in addition to the switch of the NiFe film at low field, there are switches from the Co film, first at a midfield H_{c1} , then at a higher field H_{c2} , before saturation at large fields. As we increase the value of ϕ from 15° to 45° , the height of the plateaux decreases, while H_{c2} increases, as shown in Fig. 1(b–e). When ϕ is close to 45° or the $[010]$ direction, the height of the plateaux is so small that the plateaux are almost parallel to the minimum value of GMR. With further increase of the ϕ value, the plateaux appear again in the GMR loops, as shown in Fig. 1(f). The loop shape in the region $45^\circ\text{--}90^\circ$ is nearly symmetric to that of $45^\circ\text{--}0^\circ$ except the trend of the values of H_{c2} and H_{c1} .

The reversal of 180° Co magnetization via two steps of 90° transition is termed two-jump Co magnetic switching. In Fig. 2 cartoons of two-jump Co switching (ii–iv), with different values of ϕ , are shown. At negative saturation, the spin valve is in a parallel state ($\theta = 0^\circ$), with the lowest GMR value, as the Co and NiFe magnetizations are aligned along the negative field direction. As the field strength is reduced to zero, the magnetizations undergo coherent rotation from the saturation direction and move towards the nearest easy axis, the $[\bar{1}10]$ direction in the case of $\phi = 30^\circ$. The NiFe magnetization is instantly switched and saturated along the positive field direction

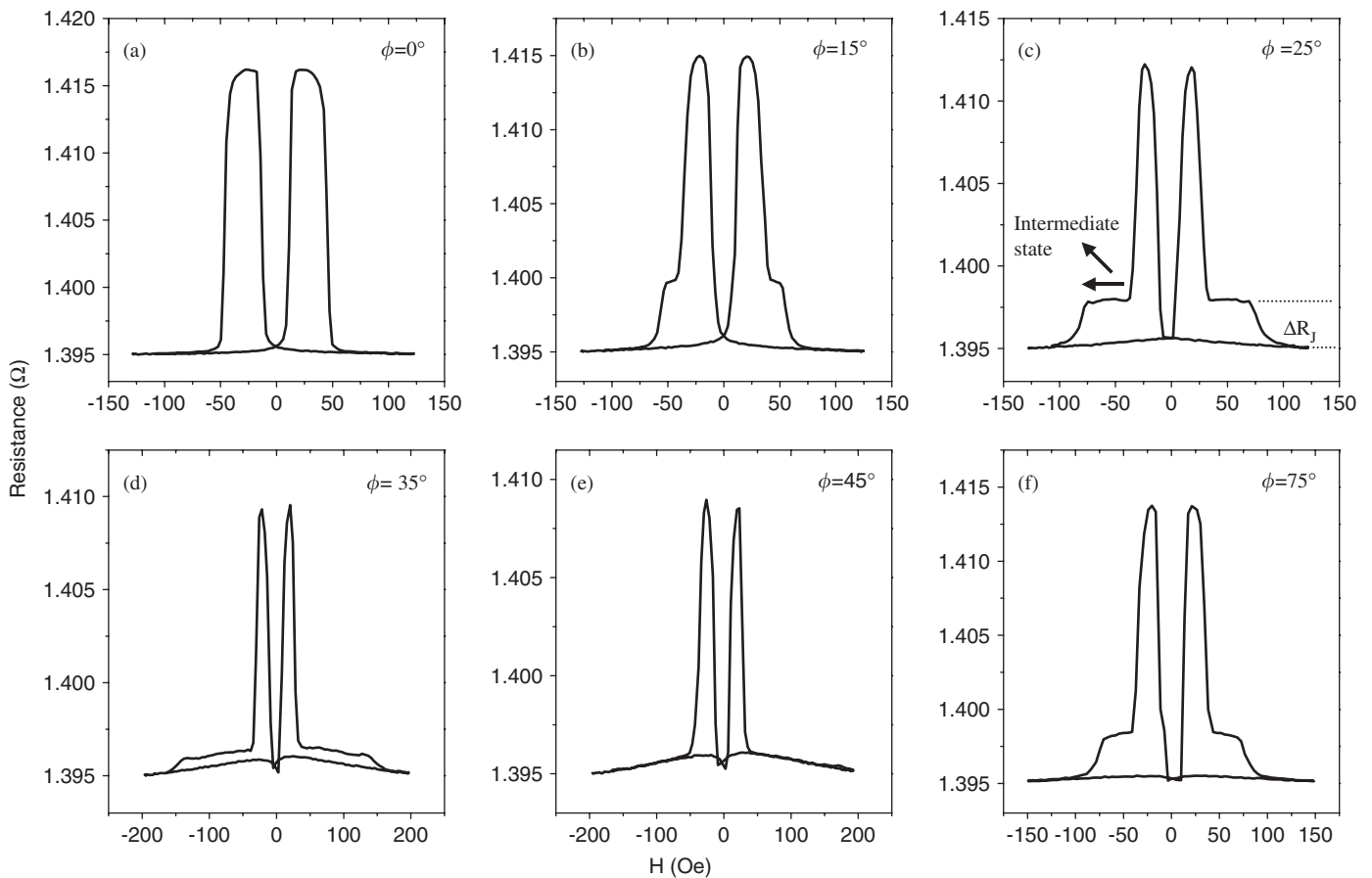


Fig. 1. Angle-dependent GMR loops showing one-jump and two-jump magnetic switching of the FCC-Co film in a NiFe/Cu/Co spin-valve structure. All measurements were carried out at room temperature. ΔR_J is the resistance difference between $R_{\text{intermediate}}$ and R_{parallel} .

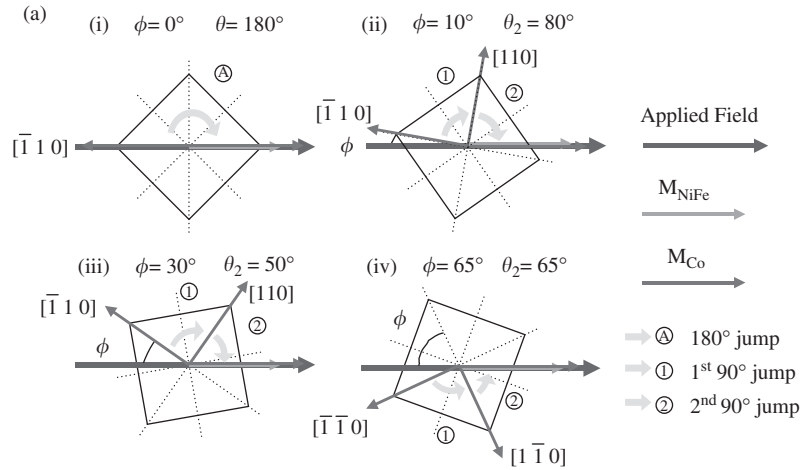


Fig. 2. Schematic of the one-jump (i) and two-jump (ii–iv) magnetic switching when the sample is rotated with an angle ϕ , where ϕ is the angle between the applied field direction and the sample $[\bar{1}10]$ axis; and θ is the angle between the Co magnetization and the NiFe magnetization, which will determine the GMR value.

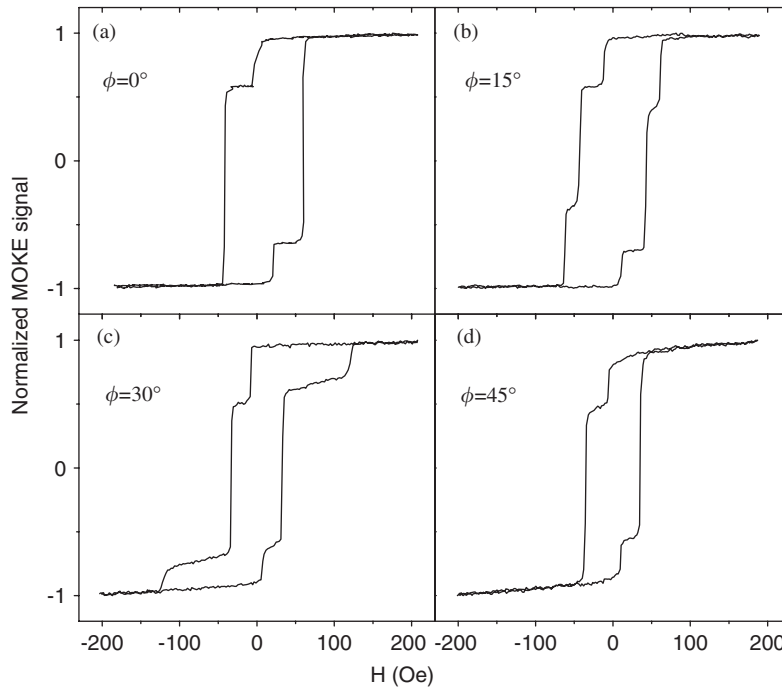


Fig. 3. Angle-dependent longitudinal MOKE hysteresis loops, showing one-jump and two-jump magnetic switching of FCC-Co in a NiFe/Cu/Co spin-valve structure.

when the applied field strength is increased slightly over the NiFe switching field. This is the antiparallel state with $\theta_1 = 150^\circ$, and the GMR has its highest value. When the field is increased in the positive direction to a value near H_{c1} , domains with 90° walls are nucleated along the $[110]$ direction. Increasing the field strength to H_{c1} allows almost $[110]$ -oriented domains to grow through wall motion process, as revealed by the sharp decrease of the GMR value. This has caused the Co first 90° jump seen in the GMR loop. When the wall motion is complete, the Co film is in single-domain state, with the direction of magnetization lying close to the $[110]$ direction. This is the new stable

intermediate state with $\theta_2 = 50^\circ$. A larger value of θ_2 corresponds to higher GMR (ΔR_j) in the spin valve sample. The Co single domain is stabilized along the $[110]$ direction, as revealed by the plateau, until the positive field strength is increased further to H_{c2} . Nucleation of 90° walls and growth of domains then quickly take place, until the film reaches the final single-domain state, in which the magnetization is uniformly oriented close to the $[1\bar{1}0]$ direction. This has caused the Co second 90° jump seen in the GMR loop. The gradual decrease of the GMR value to its minimum is due to the rotation of the Co magnetization away from the $[1\bar{1}0]$ direction towards that of the positive

applied field. In the case $45^\circ < \phi < 90^\circ$, take $\phi = 65^\circ$ for instance, the nearest easy direction to which the Co magnetization is rotated, when the field is reduced to zero, is the $[1\bar{1}0]$ direction. In the first 90° jump, the Co magnetization is at an angle $\theta = 65^\circ$ to the NiFe magnetization. Therefore, contrary to the ΔR_J behavior in the region 0° – 45° , the ΔR_J is increasing in the region from 45° to 90° .

Fig. 3 shows the measured angle-dependent LMOKE hysteresis loops of the spin-valve sample. Measurement along the $[\bar{1}10]$ axis, or 0° , reveals a double-switch

hysteresis loop (Fig. 2(a)), that corresponds to the magnetic switching of the NiFe film and the one-jump Co film. There is only one plateau, and it indicates the antiparallel state. When the sample is rotated with an angle ϕ with respect to the field orientation, triple-switch loops with two plateaux are obtained. The first switch is due to the NiFe film reversal, and the second and third switches are caused by the two-jump reversal of the Co film. In addition to the antiparallel state, an intermediate state, as explained in the GMR results, is also present, which corresponds to the second plateau. The values of H_c , H_{c1} and H_{c2} in the LMOKE hysteresis loop are generally in agreement with those in the GMR results. Therefore, the LMOKE measurements unambiguously support the GMR results.

Fig. 4 shows complete $360^\circ \pm 5^\circ$ angular coercivity data, taken from the measured LMOKE hysteresis loops. The angular coercivity results reveal fourfold coercivity symmetry of the film, which is one of the important magnetic properties of FCC-Co, and surprisingly, a weak two-fold symmetry is also present. Here, if we take coercivity as a direct consequence of magnetic anisotropy, then we can state very generally that the symmetry of the anisotropy will also be the symmetry of the coercivity. Therefore, we can conclude that, in addition to the expected cubic anisotropy, an in-plane uniaxial anisotropy oriented along the $\langle 100 \rangle$ is also found in the Co film. With regard to the two mechanisms of one-jump and two-jump switching, we see in Fig. 3 that for the Co film, fields applied at up to $\pm 10^\circ$, with $\pm 5^\circ$ error, about $[\bar{1}10]$ and $[1\bar{1}0]$ directions led to a one-jump process, and all other orientations led to a two-jump process. The angular coercivity data are fitted with a model as given in Ref. [5].

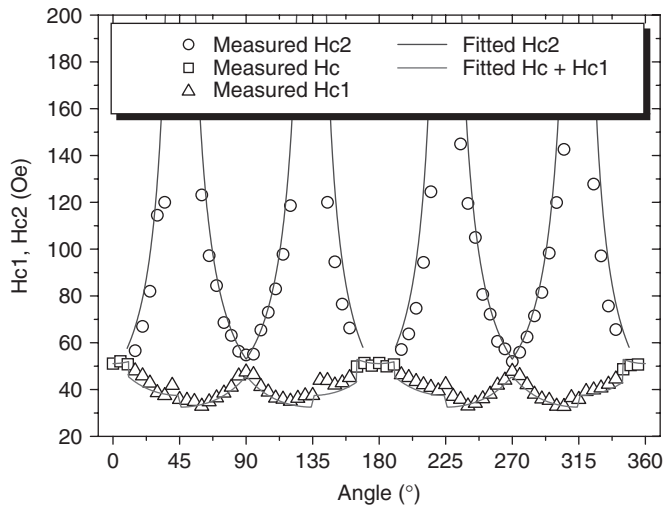


Fig. 4. The experimentally measured coercivities as a function of applied field angle. Open circles denote measured H_{c2} values; open triangles denote measured H_{c1} and open squares denote measured H_c values. The solid curves are the results of fitting using a domain-wall pinning model.

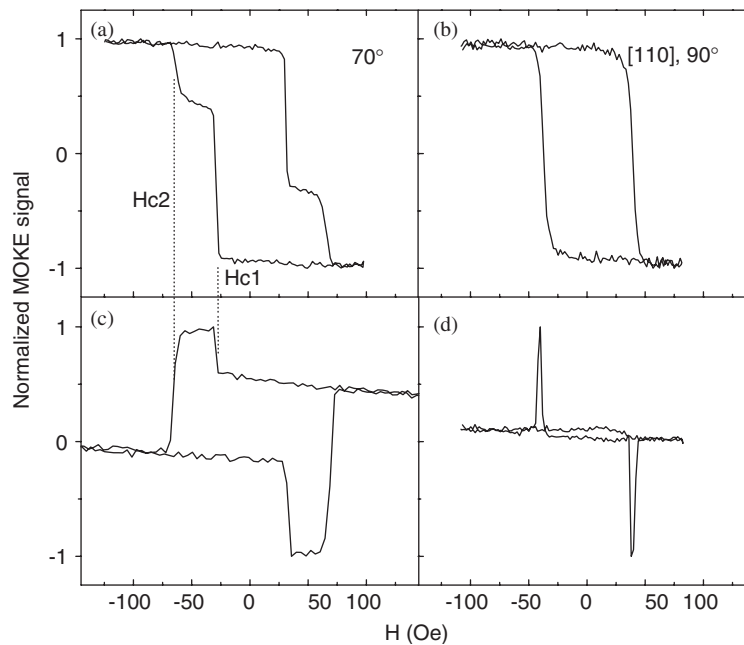


Fig. 5. (a) and (b) are the longitudinal MOKE measurements of a $\text{Si}(001)/\text{Cu}(100\text{ Å})/\text{Co}(40\text{ Å})/\text{Cu}(50\text{ Å})$ sample at 70° and 90° , respectively. Loop (c) and (d) are their transverse MOKE measurements.

To confirm the existence of the perpendicular magnetization after the first 90° jump in the Co film, transverse MOKE measurements were carried out on a Co film. The sample with a structure Si(001)/Cu(100 Å)/Co(40 Å)/Cu(50 Å), was grown in similar conditions to the spin-valve sample. Fig. 5 shows the measured LMOKE and TMOKE hysteresis loops for two different angles. Unlike the spin-valve hysteresis loops, without the presence of a NiFe film in the structure, all the switches are ascribed to the reversal process of Co film. Consider a two-jump reversal at $\phi = 70^\circ$. The LMOKE hysteresis loop has a two-step profile, with a distinct intermediate level between H_{c1} and H_{c2} . This intermediate plateau takes the form of a peak in the TMOKE loop, since the component of magnetization perpendicular to the applied field is greatest in the intermediate state. The values of H_{c1} and H_{c2} can be obtained from either type of loop. In the LMOKE loop, H_{c1} and H_{c2} are the fields marking the jump to the intermediate level, and the jump away from the intermediate level, respectively. In the TMOKE loop, H_{c1} and H_{c2} are the fields at the two sides of the intermediate peak. Note, however, that in some of the loops the two-jump behavior is less obvious. Consider the measurement taken at $[110]$ axis or $\phi = 90^\circ$. In the LMOKE loop, the intermediate plateau appears to be absent, however, the TMOKE loop still shows peaks, indicating the existence of the intermediate state. In this case, H_{c1} and H_{c2} are too close together to resolve a plateau in the LMOKE loop. However, since the TMOKE peaks are more distinct features, we can clearly identify a two-jump reversal.

In conclusion, using GMR and MOKE measurements, orientation-dependent magnetization reversal has been observed in a 40 Å epitaxial FCC-Co film. Angle-dependent measurements indicate that the reversal proceeds via one or two jumps depending on the field direction.

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